Solutions to Exploration 3.1

1. Each couple
2. Leaned right or left while kissing; categorical
3. Parameter---the true value is not observed
4. No, we don't know the exact value of pi based on the observed data. The parameter value is a fixed but unknown quantity, Our statistic from our observed data is the best guess of the value for the parameter.
5. Null: 50% of couples lean their heads to the right while kissing; Alt: More than 50% of couples lean their heads to the right while kissing
   Null: $\pi = 50\%$; Alt: $\pi > 50\%$
6. $80/124 = 0.645 = \hat{p}$
7. The p-value = 0; we have very strong evidence that the proportion of couples who lean their heads to the right while kissing is more than 50%
8. The p-value = 0.3170; we don't have evidence that the proportion of couples who lean their heads to the right while kissing is different than 60%; the null hypothesis is plausible.
9. No, the p-value is not less than 0.05. The value 0.60 is a plausible value for the probability of a randomly observed kissing couple leaning to the right.
10. We have not proven that 60% of all kissing couple lean their heads to the right; 60% is one (of many) plausible values for this probability
11. (Answers may vary a bit due to simulation)

Prob. Under Null	0.54	0.55	0.56	0.57	0.58	0.59	0.60
Two-sided p-value	0.03	0.04	0.06	.10	.12	.22	.33
Reject or plausible	Reject	Reject	Plausible	Plausible	Plausible	Plausible	Plausible

Prob. Under Null	0.70	0.71	0.72	0.73	0.74	0.75	0.76

Two-sided p-value	.22	.11	0.07	0.04	0.02	0.008	0.002
Reject or plausible	Plausible	Plausible	Plausible	Reject	Reject	Reject	Reject

12. 95% confidence interval is ~0.56 to 0.72.
13. No, the 95% confidence interval does not include 0.5, but it does include 0.6. This is in line with our answers to questions 7 and 8, which indicated that 0.6 was a plausible value but 0.5 was not
14. Wider, because we're more confident we've "caught" the correct value of the parameter in our interval
15. Needed to test probability of success $\pi = 0.52$ (p-value = 0.004), $\pi = 0.53$ (p-value = 0.012), to determine that the interval of plausible values at 0.01 significance is 0.53 to 0.74
16. 99% confident
17. 99% confidence interval is wider. Yes, is consistent with previous prediction.
18. 90% will be narrower than both the 95% and 99%. All other things being equal, as the level of confidence increases, so does the width of the confidence interval.

Solutions to Exploration 3.2

1. Population: all US adults, Sample: 1,019 US adults

2. Yes, it was a random sample.

3. 80% is based on the sample, not the population. $\hat{p}$

4. The proportion of all US adults who agreed with the statement

5. No, the proportion of all adult Americans is likely not 80% (we only talked to 1,019), but is likely close to 80%

6. The p-value is approximately 0.048. Reject this value at the 0.05 significance level.

7. Mean = 0.775, SD = 0.013

8. The p-value of approximately 0. 50% is not a plausible value for the proportion of all US adults who agreed with the statement; Reject 50% at the 0.05 significance level. Mean = 0.50, SD = 0.016.

9. (0.80-0.775)/0.013 = 1.92. So, 0.80 is 1.92 standard deviations from 0.775

10. Similar, a bit larger when using 0.50

11. a. $2 \times 0.016 = 0.032$ b. 0.80 +/- 0.032 = (0.768, 0.832) c. We are 95% confident that the proportion of all US adults that agree with the statement is between 0.768 and 0.832.

12. $\sqrt{\hat{p}(1-\hat{p})/n} = \sqrt{0.80(1-0.80)/1019} = 0.013$. Very similar to what we got for the standard deviation from simulation

13. (0.7754, 0.8246)

14. Yes, very similar

15. 99% CI: (0.7677, 0.8323). Midpoint is the same; interval is wider because margin of error increased.

Solutions to Exploration 3.3

1. Answers will vary with class data. One example follows: $n = 22$, Sample mean = 6.705, sample SD = 1.297.
2. 6.705, since that is the sample mean.
3. No, every sample will yield a different estimate of the true (population) mean.
4. Got approximately 0.32 when simulating. This value (SD of sample means) is less than SD of the data.
5. 1.297/sqrt(22) = 0.28
6. 6.705 +/- 2 × 0.28 = (6.1, 7.3) We are 95% confident that the average hours of sleep per night all students at your college got last night was between 6.1 and 7.3.
7. No, the 95% confidence interval does not contain the hypothesized 8 hour average under the null. We already knew this because the two-sided p-value was much smaller than the significance level of 0.05.
8. (6.1299, 7.2801) We are 95% confident that the average hours of sleep per night all students at your college got last night was between 6.13 and 7.28. Yes, they are essentially the same. Could have anticipated this because validity conditions are met.
9. Yes, the sample size was at least 20 and not strongly skewed.

Solutions to Exploration 3.4A

1. Obs units: 1,039 US adults, variable: How much they plan to spend on Christmas presents, a quantitative variable
2. The average amount all US adults plan to spend on Christmas presents
3. Sample size (n) = 1,039; sample mean ($\bar{x}$) = \$704.
4. The sample standard deviation is approximately equal to the average deviation (distance) of individuals' Christmas spending amounts from the mean of \$704.
5. 704 +/- 2 × 150/sqrt(1,039) = 704 +/- \$9.3 = (\$694.7, \$713.3)
6. 704 +/- \$18.6. The interval is twice as wide as the one from #5,which makes sense because s is twice as large. The interval is centered at the same value (\$704)
7. Larger, because the sample size is smaller.
8. The ME is 2 × 150/sqrt(477) = \$13.74. The ME is larger as predicted.
9. As the sample size increases, the width of the confidence interval decreases. With increased sample size comes increased precision of the estimate.
10. Of the 86 responses reported in the table (only a fraction of the total sample), a standard deviation of 300 seems more reasonable because too large a proportion of responses are more than 1 std dev from the mean if the std dev is only 150. If the std dev is 150, then 26 + 16 + 19 + 4 = 65 out of 86 responses would be more than 1 std dev from the mean, which is too many. 300 is a better estimate of the standard deviation.
11. While the sample size is large, the data looks to be fairly skewed and so even with a sample size of greater than 20 we may need to be cautious.

Solutions to Exploration 3.4B

1. No.
2. No.
3. No. No.
4. (answers will vary) a. p-hat = 0.510, b. (0.480, 0.541), c. Yes, it is always the midpoint. d. Yes, because we're 95% sure the interval will "catch" the true parameter value.
5. a. 0.50. Yes, on average the center of the intervals is 0.50, which is the long-run probability value. b. 0.466, (0.436, 0.497), the sample proportion is in the interval, but the long-run probability (0.50) is not, 0.504, (0.474, 0.535), yes, sample proportion is in the interval, and the long-run probability (0.50) is as well. c. 96/100 = 96%. d. Running it forever would get to 95%. e. 95%, 5%

6. a. i. Width will be more narrow for 90% confidence interval compared to the 95% confidence interval. ii. 90% will be green and 10% will be red in the long run. b. i. More narrow, not quite as sure interval "catches" the long-run probability. ii. Running total is closer to 90% c. i. Widths will be more narrow. ii. The % of successful (green) intervals will stay at 90%. d. i. more narrow, larger sample is more precise. ii. no change in running total %, level of confidence hasn't changed. e. i. Incorrect; this seems as if the long-run probability is bouncing around, not fixed ii. Incorrect; the endpoints should be.461 and .589 iii. Correct; this is the definition of a confidence interval.

Solutions to Exploration 3.5A

1. $0.68 = \hat{p}$
2. The proportion of all American adults who claim to have voted in the 1996 Presidential election.
3. (0.6589, 0.7059)
4. No, only those that claimed to have voted, not those who actually voted.
5. No.
6. The p-value would be less than 0.01.
7. All American adults is the population, 2.4 million responders is the sample. The percent of American adults who will vote for Republican Alf Landon is the parameter and is unknown. The proportion of the 2.4 million who say they will vote for Republican Alf Landon is the statistic and is known to be 0.57.
8. Even though the level of confidence is high, the sample size is quite large and so we expect the interval to be narrow.
9. 0.57, m.e. is 0.0003 , (0.5697, 0.5703)
10. Yes.
11. No. No.
12. Selecting a non-representative sample. The sampling frame used only polled those who had cars or phones or were subscribers to the magazine. These people would be more affluent and traditionally more affluent voters are Republicans.
13. Gallup used a non-biased sampling method.

Solutions to Exploration 3.5B

1. Statistic, $\hat{p}$
2. The proportion of all American households in 2007 that own a pet cat, π.
3. Null: $\pi = 33.3\%$; Alt: $\pi < 33.3\%$; $z = -4.14$; approximate p-value = 0; the result is statistically significant at the 0.001 level. We have very strong evidence against the null and in favor of the alternative that significantly less than 1/3 of all American households in 2007 owned a pet cat. We know this because the probability of getting our study result or something even less assuming 1/3 of all households own a pet cat, is essentially zero.
4. (0.3169, 0.3311)
5. The confidence interval is so narrow because the sample size is so large.
6. The first is much evidence against the value under the null hypothesis being the value of the parameter; the second is stating that the true parameter value is much different than the value under the null hypothesis.
7. First uses a p-value; second uses a confidence interval.
8. The first, but not the second since the confidence interval is very close to 1/3.
9. 0.25 or lower
10. No.
11. 0.161
12. No.
13. 0.161
14. Increasing the significance level increases power. This makes it easier to find a significant result and power is the probability of finding a significant result when you are supposed to.
15. 0.27 or lower. 0.291 power increased.
16. Increasing the sample size increases power. This makes the standard deviation of the null distribution smaller, so it is easier to detect differences from the hypothesized null value and thus easier to find a significant result.
17. 0.29 or lower. 0.34 power increased.
18. Decreasing the actual population value would increase the power since it would be further away from the hypothesized value, making it easier to find a significant result.
19. 0.91. Power increased.